

Higher-Order Metaphysics

Week Three: Prior's Theorem, the Russell-Myhill Theorem, and the debate about Beta

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1 Recap: H and H_0

2 Structure-theoretic objections to β -conversion

A natural way filling in the “structure”-theoretic objection to β -conversion will appeal to one or both of the following principles.

Predicate structure $\forall XX'yy'.Xy =_{\tau} X'y' \rightarrow X =_{\sigma \rightarrow t} X'$

Argument structure $\forall XX'yy'.Xy =_{\tau} X'y' \rightarrow y =_{\sigma} y'$

β -conversion conflicts with both. Using $\forall$ Elim, instantiate X with $\lambda z.Rzb$, X' with $\lambda z.Raz$, y with a and y' with b . Then $Xy = (\lambda z.Rzb)a \sim_{\beta} Rab \sim_{\beta} (\lambda z.Raz)b = X'Y'$, so the antecedent of the two principles becomes a theorem (provable using $=$ Intro and β -conversion). So Argument Structure implies the absurd $a =_{\sigma} b$, and Predicate Structure implies the absurd $(\lambda z.Rzb) =_{\sigma \rightarrow \tau} (\lambda z.Raz)$, both of which will be obviously false for many cases.

3 Prior's theorem

Argument 1: Suppose the person in room 7 is asserting that everything the person in room 7 is asserting is false. Then, if everything the person in room 7 is asserting is false, something the person in room 7 is asserting is true—a contradiction. So, not everything the person in room 7 is asserting is false. So, something the person in room 7 is asserting (viz., that everything the person in room 7 is asserting is false) is false, although not every such thing is false. So the person in room 7 is asserting at least two things.

Argument 2: Suppose Sue is contemplating the property of lacking all the properties one is contemplating. Then, if Sue lacks all the properties she is contemplating, she has some property she is contemplating—a contradiction. So, she does not lack all the properties she is contemplating. So, she lacks some property she is contemplating (viz., that of lacking all the properties one is contemplating), although she does not lack every such property. So she is contemplating at least two properties.

To formalize the structure common to these two arguments, let $a_1, \dots, a_n$ (for short, $\vec{a}$) be zero or more terms of types $\sigma_1, \dots, \sigma_n$ (for short, $\vec{\sigma}$).¹ Let Q be a constant or variable of type $(\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t) \rightarrow \sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t$ (for short, $(\vec{\sigma} \rightarrow t) \rightarrow \vec{\sigma} \rightarrow t$).² Let $\vec{x}$ be variables of the same types as $\vec{a}$, and Y a variable of type $\vec{\sigma} \rightarrow t$.³

¹ In the first instantiation above $n = 0$; in the second instantiation, $n = 1$, σ_1 is e , and a_1 is ‘Sue’.

² In the first instantiation above Q is of type $t \rightarrow t$, interpreted as ‘is being asserted by the person in room 7’; in the second instantiation, Q is of type $(e \rightarrow t) \rightarrow e \rightarrow t$, interpreted as ‘is being contemplated by’.

³ Note: when $n = 0$, we understand $\lambda\vec{x}.A$ as just A .

1	$Q(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a}$	
2	$\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a}$	
3a	$(\lambda Y.QY\vec{a} \rightarrow \neg Y\vec{a})(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})$	$\forall E, 2$
3b	$Q(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a} \rightarrow \neg(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a}$	$E\beta, 3a$
3	$Q(\lambda\vec{x}.\forall p.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a} \rightarrow \neg\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a}$	$E\beta, 3b$
4	$\neg\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a}$	$\Rightarrow E, 3, 1$
5	$\perp$	$\neg E, 4, 2$
6	$\neg\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a}$	$\neg I, 2-5$
7	$Q(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a} \wedge \neg\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a}$	$\wedge I, 1, 6$
7a	$Q(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a} \wedge (\lambda\vec{x}.\neg\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a}$	$E\beta, 7$
7b	$(\lambda Y.QY\vec{a} \wedge \neg Y\vec{a})(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})$	$E\beta, 7a$
8	$\exists Y.QY\vec{a} \wedge \neg Y\vec{a}$	$\exists I, 7b$
9	$(\exists Y.QY\vec{a} \wedge \neg Y\vec{a}) \wedge (\neg\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a})$	$\wedge I, 8, 6$
10	$Q(\lambda\vec{x}.\forall Y.QY\vec{x} \rightarrow \neg Y\vec{x})\vec{a} \rightarrow (\exists Y.QY\vec{a} \wedge \neg Y\vec{a}) \wedge (\neg\forall Y.QY\vec{a} \rightarrow \neg Y\vec{a})$	$\Rightarrow E, 1-9$

We could if we wished tack on a proof from line 9 of $\neg\forall YZ.QY\vec{a} \wedge QZ\vec{a} \rightarrow Y = Z$ (‘It’s not the case that Sue is contemplating at most one property’) or $\neg\forall YZ.QY\vec{a} \wedge QZ\vec{a} \rightarrow \forall\vec{x}.Y\vec{x} \leftrightarrow Z\vec{x}$ (‘It’s not the case that all the properties Sue is contemplating are coextensive’), or (using DNE) $\exists YZ.QY\vec{a} \wedge QZ\vec{a} \wedge Y \neq Z$ (‘Sue is contemplating at least two properties’); but the above is sufficiently surprising.⁴

⁴ Exercise: write up one of these further derivations (either natural deduction or axiomatic).

- Note that the above derivation will still be valid if we pick any formula P and variables $Z : \vec{\sigma} \rightarrow t$ and $\vec{u} : \vec{\sigma}$ and replace each formula of the form $QA\vec{b}$ with $P[A/Z, \vec{b}/\vec{u}]$. Given β -conversion this extra generality isn’t relevant since we can recover the substitution version of the theorem from the one above by taking Q to be $\lambda Z\vec{u}.P$. But if one rejects β , the substitution-theoretic version of the theorem-schema is more general.

4 Existential variant of the theorem

Suppose Max is contemplating the property of lacking some property one is contemplating. Then, if Max has every property he is contemplating, Max lacks some property he is contemplating—a contradiction. So, Max lacks some property he is contemplating. So, Max has some property he is contemplating (viz., lacking some property one is contemplating), although he does not have every property he is contemplating. So Max is contemplating at least two propositions.

1		$Q(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a}$	
2		$\forall Y. QY\vec{a} \rightarrow Y\vec{a}$	
3a		$(\lambda Y. QY\vec{a} \rightarrow Y\vec{a})(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})$	$\forall E, 2$
3b		$Q(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a} \rightarrow (\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a}$	$E\beta, 3a$
3		$Q(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a} \rightarrow \neg \forall Y. QY\vec{a} \rightarrow Y\vec{a}$	$E\beta, 3b$
4		$\neg \forall Y. QY\vec{a} \rightarrow Y\vec{a}$	$\Rightarrow E, 3, 1$
5		$\perp$	$\neg E, 4, 2$
6		$\neg \forall Y. QY\vec{a} \rightarrow Y\vec{a}$	$\neg I, 2-5$
7		$Q(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a} \wedge \neg \forall Y. QY\vec{a} \rightarrow Y\vec{a}$	$\wedge I, 1, 6$
7a		$Q(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a} \wedge (\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a}$	$E\beta, 7$
7b		$(\lambda Y. QY\vec{a} \wedge Y\vec{a})(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})$	$E\beta, 7a$
8		$\exists Y. QY\vec{a} \wedge Y\vec{a}$	$\exists I, 7b$
9		$(\exists Y. QY\vec{a} \wedge Y\vec{a}) \wedge (\neg \forall Y. QY\vec{a} \rightarrow Y\vec{a})$	$\wedge I, 8, 6$
10		$Q(\lambda \vec{x}. \neg \forall Y. QY\vec{x} \rightarrow Y\vec{x})\vec{a} \rightarrow (\exists Y. QY\vec{a} \wedge Y\vec{a}) \wedge (\neg \forall Y. QY\vec{a} \rightarrow Y\vec{a})$	$\Rightarrow E, 1-9$

5 Learning to live with the theorem

Here are two particularly challenging instances of the theorem.

- If the person in room 7 seems to be asserting that there's something false that the person in room 7 seems to be asserting, then there are at least two propositions (one true, one false) that the person in room 7 seems to be asserting.
- If 'is something that lacks every property it expresses' expresses the property of being something that lacks every property it expresses,

then ‘is something that lacks every property it expresses’ expresses at least two (non-coextensive) properties.

We can distinguish two general strategies: *keep the appearances, give up uniqueness* (me) and *keep uniqueness, give up the appearances* (Prior, Bacon).

Both come in a *radical* form and a *sensitive* form.

- Radical anti-uniqueness: ordinary, unproblematic utterances (e.g. an utterance of ‘Everything being asserted by the person in room 7 is false’ in some room other than 7) routinely involve the assertion of multiple things; ordinary, unproblematic expressions routinely express multiple things...
 - Sensitive anti-uniqueness: Uniqueness fails only in extraordinary cases—e.g. it would not fail in the case of an utterance of ‘Everything being asserted by the person in room 7 is false’ in some room other than 8. So there is a surprising new kind of “externalism” where intuitively irrelevant factors (e.g. the number on the door of the room where an utterance is being made) make a difference to what’s being asserted by the utterance.
- (i) Radical appearance-rejection (Bacon): the surprises are everywhere—e.g., an English-speaker who uttered ‘Everything being asserted by the person in room 7 is false’ in some room other than 7 would (probably) not thereby assert that everything being asserted by the person in room 7 is false. And it would also (probably) not seem to anyone that they had done so...
- (ii) Sensitive appearance-rejection (Prior): an English-speaker an English-speaker who uttered ‘Everything being asserted by the person in room 7 is false’ in some room other than 7 would (probably) thereby assert that everything being asserted by the person in room 7 is false. So, there is a surprising new kind of “externalism” where intuitively irrelevant factors (e.g. the number on the door of the room where an utterance is being made) make a difference to what (if anything) is asserted by it.

6 The Russell-Myhill theorem

For the Russell-Myhill theorem, let $n = 1$ and $\sigma_1 = t$. Choose some arbitrary sentence r . Let $Q : (t \rightarrow t) \rightarrow t \rightarrow t$ be $(\lambda Y^{t \rightarrow t} p^t. p =_t Yr)$, and let $a : t$ be $(\lambda x. \forall Y. QYx \rightarrow \neg Yx)r$.

Then the antecedent $Q(\lambda x. \forall Y. QYx \rightarrow \neg Yx)((\lambda x. \forall Y. QYx \rightarrow \neg Yx)r)$ is immediately β -equivalent to

$$(\lambda x. \forall Y. QYx \rightarrow \neg Yx)r =_t (\lambda x. \neg \forall Y. QYx \rightarrow \neg Yx)r$$

and is thus itself a theorem of H_0 (by $=\text{Intro}$ and $E\beta$). So the consequent is also a theorem. It is logically equivalent by $E\beta$ to:

$$(\exists Y. a = Yr \wedge \neg Ya) \wedge (\neg \forall Y. a = Yr \rightarrow \neg Ya)$$

This implies

$$\exists Y Y'. Y \neq_{t \rightarrow t} Y' \wedge Yr =_t Y'r$$

which contradicts Predicate Structure.

We can run a parallel argument against Argument structure by choosing some arbitrary term $R : (t \rightarrow t) \rightarrow t$ —e.g. R could be $\forall_t$, or $\lambda X^{t \rightarrow t}. Xr$ —and letting Q be $(\lambda Y^{t \rightarrow t} p^t. p =_t RY)$ and a be $R(\lambda x. \forall Y. QYx \rightarrow \neg Yx)$. Then we'll derive

$$(\exists Y. a = RY \wedge \neg Ya) \wedge (\neg \forall Y. a = RY \rightarrow \neg Ya)$$

and hence

$$\exists Y Y'. Y \neq_{t \rightarrow t} Y' \wedge RY =_t RY'$$

7 Resisting by giving up propositional logic

8 Resisting by giving up the quantifier rules

Free higher-order logic gives up UI ($\forall\text{Elim}/\exists\text{Intro}$), and replaces it with the following weaker axioms (all theorems of H_0):

Free Instantiation	$\forall x(\forall F \rightarrow Fx)$
Quantifier Normality	$(\forall x. P \rightarrow Q) \rightarrow (\forall x. P) \rightarrow (\forall x. Q)$
Quantifier Exchange	$(\forall xy. P) \rightarrow (\forall yx. P)$
Quantifier Duality	$(\forall x. \neg P) \leftrightarrow (\neg \exists x. P)$

This however yields a *very* weak system, consistent with $\forall x^\sigma. \perp$ in each type. Proponents face the challenge of finding further principles which can restore a bit more of the standard theoretical role of quantification.

9 Comprehension

Here is one principle that could help play this role:⁵

$$\text{Comprehension} \quad \exists X^{\vec{\sigma} \rightarrow t}. \forall y^{\vec{\sigma}}. X\vec{y} \leftrightarrow P$$

⁵ This is a theorem of H_0 : instantiate X with $\lambda \vec{y}. P$.

This does not bring us back to classical logic, but does give us limited versions of the quantifier rules restricted to terms that are provably *extensional in their first argument place*:

$$\frac{\Gamma \vdash \forall_{\vec{\sigma} \rightarrow t} F \quad \vdash (\forall \vec{z}. X\vec{z} \leftrightarrow Y\vec{z}) \rightarrow (FX \rightarrow FY)}{\Gamma \vdash FA} \text{ } \forall\text{Elim}_{\text{ext}} \quad \frac{\Gamma \vdash FA \quad \vdash (\forall \vec{z}. X\vec{z} \leftrightarrow Y\vec{z}) \rightarrow (FX \rightarrow FY)}{\Gamma \vdash \exists_{\vec{\sigma} \rightarrow t} F} \text{ } \exists\text{Intro}_{\text{ext}}$$

Using these restricted rules, the proof of Prior's theorem will still go through provided we choose Q to be something *extensional in its first argument place*, $\vdash (X\vec{z} \leftrightarrow Y\vec{z}) \rightarrow (QX\vec{x} \rightarrow QY\vec{x})$.

This is enough to run a variant of the Russell-Myhill theorem. As before, choose an arbitrary r . Let $Q : (t \rightarrow t) \rightarrow t \rightarrow t$ now be $\lambda Y p. \exists Z. Y \equiv Z \wedge p = Zr$.⁶ As before, let $a : t$ be $(\lambda x. \forall Y. QYx \rightarrow \neg Yx)r$. This Q is provably extensional in its first argument place, since

⁶ $\equiv$ abbreviates coextensionality,
 $\lambda YZ. \forall p. Yp \leftrightarrow Zp$

$$\vdash \forall Y Y'. Y \equiv Y' \rightarrow (\exists Z. Y \equiv Z \wedge p = Zr) \rightarrow (\exists Z. Y' \equiv Z \wedge p = Zr)$$

So, the proof goes through with Comprehension. The antecedent is a theorem for similar reasons to the ones we had before. And the consequent entails that there exist non-coextensive Y and Y' that both bear Q to a , i.e. such that there exist Z coextensive with Y and Z' coextensive with Y' such that a predicates both Z and Z' of r .

This is still inconsistent with Predicate Structure!

10 Bacon's argument for β -conversion

Bacon's thought: a lambda-term $\lambda v. A$ is *stipulated* to be such that, if it's meaningful at all, any expression of the form $(\lambda v. A)B$ is *synonymous* with $A[B/v]$.

Someone could perfectly well doubt that, e.g., $\exists X^{e \rightarrow t}. Rab =_t Xa$. But such a person should think that $\lambda x. Rxb$ is a meaningless term, since if it were meaningful, $(\lambda x. Rxb)a$ would be synonymous with Rab , so $Rab =_t (\lambda x. Rxb)b$ would be true, so $\exists X^{e \rightarrow t}. Rab =_t Xa$ would be true.

This is not an argument for the truth of $\exists X^{e \rightarrow t}. Rab =_t Xa$; the only argument suggested for that (and hence for the meaningfulness of the relevant λ -term) is some kind of appeal to overall theoretical virtues.

- Suppose you didn't accept that $\exists X^{e \rightarrow t}. Rab =_t Xa$. Is Bacon suggesting that you should think that *people who do accept this* (along with the rest of H) don't mean anything by $\lambda x. Rxb$? This seems drastic!

11 The argument from explicit definitions